

Lab 17: Expectation-Maximization (EM) for Gaussian Mixture Model

Aim

To implement the **Expectation-Maximization (EM) algorithm** using a **Gaussian Mixture Model (GMM)** for clustering data.

Algorithm

1. Import the required libraries.
2. Create a sample dataset with multiple clusters.
3. Initialize the Gaussian Mixture Model.
4. Fit the GMM to the dataset using the EM algorithm.
5. Predict the cluster for each data point.
6. Display the cluster labels.
7. Visualize the clustered data.

Program

```
# Lab 17: Expectation-Maximization (EM) for Gaussian Mixture Model
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.mixture import GaussianMixture
```

```
# Create sample data
```

```
np.random.seed(42)
```

```
cluster1 = np.random.normal(loc=2, scale=0.5, size=(50, 2))
```

```
cluster2 = np.random.normal(loc=6, scale=0.5, size=(50, 2))
```

```
X = np.vstack((cluster1, cluster2))
```

```
# Create Gaussian Mixture Model

gmm = GaussianMixture(
    n_components=2,
    random_state=42
)

# Train the model
gmm.fit(X)

# Predict clusters
labels = gmm.predict(X)

print("Cluster Labels:")
print(labels)

# Visualize clusters
plt.scatter(X[:, 0], X[:, 1], c=labels)

plt.title("Gaussian Mixture Model Clustering")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")

plt.show()
```

Sample Output

Cluster Labels:

```
[0 0 0 0 ... 1 1 1 1 1]
```

A scatter plot will also be displayed showing the data divided into **two clusters**.

Result

Thus, the **Expectation-Maximization algorithm** was successfully implemented using a **Gaussian Mixture Model**, and the given data was successfully clustered into two groups.

